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REVIEWER ACTIVITIES: NAVY-AS, DLA-IS

USER ACTIVITIES: NAVY-MC

DD FORM 1673-1 (OBSOLETE)

PROPOSED REVISION 5310-0947

PREPARING ACTIVITY: NAVY-OS, ARMY-AT, AIR FORCE-11

PROCUREMENT SPECIFICATION: MIL-W-21338

SUPERSEDES: MS172201 thru MS172235

INTERNATIONAL INTEREST

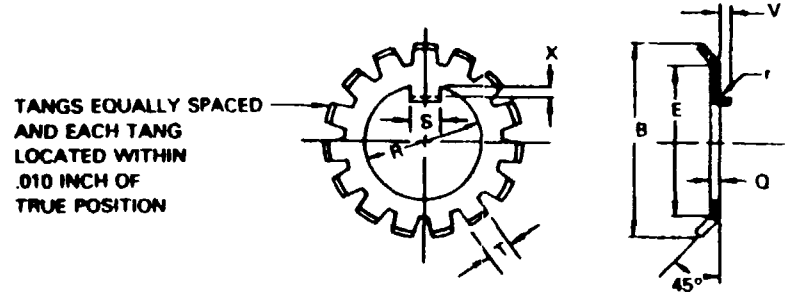
TITLE: WASHER, KEY, RETAINING, BALL AND ROLLER BEARING, LIGHT SERIES

MILITARY STANDARD

MS 19069

SHEET 1 OF 2

FSC 3110



DASH NO.	Q THICKNESS		R BORE		E +.020 −.000	B +.000 ±.030	KEY							TANGS			Ⓐ
							S		X		V		r RAD	NO.	T		
	Q	TOL ±	MAX	MIN			MAX	MIN	MAX	MIN	V	TOL ±			MAX	MIN	
A00	.042	.004	.421	.406	.531	.828	.120	.110	.030	.020	.062	.031	.016	9	.104	.084	
A01	.042	.004	.499	.484	.625	.922	.120	.110	.030	.020	.062	.031	.016	9	.104	.084	
A02	.042	.004	.616	.601	.750	1.094	.120	.110	.030	.020	.062	.031	.016	11	.104	.084	
A03	.042	.004	.694	.679	.875	1.203	.120	.110	.030	.020	.062	.031	.016	11	.104	.084	
A04	.042	.004	.816	.801	.969	1.344	.176	.156	.030	.020	.062	.031	.016	11	.166	.146	
A23	.044	.036	.816	.801	1.000	1.375	.120	.110	.030	.020	.062	.031	.016	11	.036	.034	
A05	.050	.004	1.009	.989	1.188	1.562	.176	.156	.030	.020	.094	.031	.016	13	.166	.146	
A06	.050	.004	1.213	1.193	1.438	1.859	.176	.156	.050	.035	.094	.031	.031	13	.166	.146	
A07	.050	.004	1.416	1.396	1.656	2.078	.176	.156	.050	.035	.094	.031	.031	15	.166	.146	
A08	.058	.005	1.603	1.583	1.844	2.250	.300	.280	.050	.035	.094	.031	.031	15	.234	.204	
A09	.058	.005	1.817	1.792	2.062	2.500	.300	.280	.050	.035	.125	.031	.031	17	.234	.204	
A10	.058	.005	2.017	1.992	2.250	2.688	.300	.280	.050	.035	.125	.031	.031	17	.234	.204	
A11	.063	.005	2.207	2.182	2.500	2.953	.300	.280	.050	.035	.125	.031	.031	17	.234	.204	
A12	.063	.005	2.425	2.400	2.688	3.188	.300	.280	.070	.055	.125	.031	.047	17	.234	.204	
A13	.063	.005	2.613	2.588	2.875	3.375	.300	.280	.070	.055	.125	.031	.047	19	.234	.204	
A14	.063	.005	2.816	2.791	3.094	3.594	.300	.280	.070	.055	.188	.062	.047	19	.234	.204	
A15	.063	.005	3.003	2.973	3.375	3.922	.300	.280	.070	.055	.188	.062	.047	19	.327	.297	
A16	.063	.005	3.207	3.177	3.594	4.141	.360	.340	.070	.055	.188	.062	.047	19	.327	.297	
A17	.063	.005	3.425	3.395	3.812	4.359	.360	.340	.070	.055	.188	.062	.047	19	.327	.297	
*A18	.063	.005	3.612	3.582	4.000	4.547	.360	.340	.070	.055	.188	.062	.047	19	.327	.297	
*A19	.063	.005	3.830	3.800	4.219	4.812	.360	.340	.070	.055	.188	.062	.047	19	.327	.297	
A20	.063	.005	4.018	3.988	4.406	5.000	.360	.340	.085	.065	.250	.062	.047	19	.327	.297	
*A21	.063	.005	4.222	4.192	4.594	5.188	.360	.340	.085	.065	.250	.062	.047	19	.390	.360	
A22	.063	.005	4.425	4.395	4.812	5.406	.360	.340	.085	.065	.250	.062	.047	19	.390	.360	

Ⓐ *Inactive for new design after 1987

For NOTES see sheet 2. Ⓐ denotes changes.

APPROVED 7 JUN 1961 REVISED Ⓐ 7 October 1987

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Ⓐ SUBSTITUTION TABLE

Superseded number	Replacement number	Superseded number	Replacement number
MS 172201	MS 19069-A00	MS 172213	MS 19069-A12
MS 172202	MS 19069-A01	MS 172214	MS 19069-A13
MS 172203	MS 19069-A02	MS 172215	MS 19069-A14
MS 172204	MS 19069-A03	MS 172216	MS 19069-A15
MS 172205	MS 19069-A04	MS 172217	MS 19069-A16
MS 172206	MS 19069-A05	MS 172218	MS 19069-A17
MS 172207	MS 19069-A06	MS 172219	MS 19069-A18
MS 172208	MS 19069-A07	MS 172220	MS 19069-A19
MS 172209	MS 19069-A08	MS 172221	MS 19069-A20
MS 172210	MS 19069-A09	MS 172222	MS 19069-A21
MS 172211	MS 19069-A10	MS 172223	MS 19069-A22
MS 172212	MS 19069-A11	MS 172224	MS 19069-A23

NOTES:

Ⓐ 1. MATERIAL: Alloy steel AMS-6350.

2. TEMPER: Each washer shall be of such temper as to permit its being bent 180°, without fracture, either way of the grain, over a radius equal to one half the stock thickness.

3. PROTECTIVE COATING: Washer shall be phosphate coated in accordance with DOD-P-16232, Type Z, Class 2.

4. DIMENSIONS: All dimensions are in inches. Tolerances: decimals ±.010, angles ±2° unless otherwise specified.

Ⓐ 5. PART NUMBER: The MS part number is the MS number, plus the dash number.

Example: MS19069-A00
dash number

6. NUTS: See MS19067 for applicable nuts. Applicable nuts should have dash numbers corresponding to washer numbers: e.g., washer size A00 should be used with nut size number A00.

7. OTHER NOTES:
(a) In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
(b) Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

USER ACTIVITIES

REVIEWER ACTIVITIES

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APPROVED 7 JUNE 1961 REVISED Ⓐ For changes see sheets 1 and 2

P A	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY-OS (Other Cog) ARMY-AT AIR FORCE-11		WASHER, KEY, RETAINING, BALL AND ROLLER BEARING, LIGHT SERIES	MS 19069
PROCUREMENT SPECIFICATION MIL-W-21338	SUPERSEDS MS 17200 thru MS 17224		SHEET 2 OF 2

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DD FORM 672-1 COORDINATED

PROPOSED REVISION 5310-0947